Thakaa at AraGenre 2026: Definition-Guided LLM Judging with Full-Context Multi-Model Consensus for Hierarchical Arabic Genre Classification

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Abstract

We report the first-place system in AraGenre 2026 (0.7352 Hierarchical Macro F1, 18 teams). The task asks for one of 6 broad and one of 74 fine-grained specific genres for each of 27,972 hidden-test Arabic segments, with every genre given only as an English prose definition and the specific-label inventory *disjoint* from the released training data. Our pipeline is training-free: it makes no gradient updates on the task, though the judge prompt does carry hand-written per-genre cues. A family-restricted, definition-guided LLM judge scores candidate genres setwise under a hierarchically constrained decoding scheme, reaching 0.7139. The public scoreboard showed that this base already led on Specific Macro F1 but trailed on Broad Macro F1, so we target the broad tier with a full-context multi-model consensus stage: two instruction models each see the *entire* 74-definition taxonomy and re-predict the specific genre, and a correction is applied only where both agree against the base. This lifts Broad Macro F1 from 0.879 to 0.892 by suppressing a topic trap in which a book *about* religion is taken for religious scripture. A three-arm ablation separates the trap's two causes: an explicit type-versus-topic instruction removes 41% of the trapped calls (394 $\rightarrow$ 233), and showing the full taxonomy removes a further 49% (233 $\rightarrow$ 119). A final attractor drain redistributes over-predicted generic classes to starved rare siblings, carrying the base from 0.7139 to the submitted 0.7352.

1 Introduction

Genre is what a text *does* rather than what it is about: the same subject appears in a contract, a news report and a textbook, and only the first is useful to someone assembling a legal corpus. Sorting documents this way is what makes an archive filterable and a corpus assemblable by register, and Arabic has far less of this work behind it than English. AraGenre 2026 (El-Haj et al., 2026) frames the problem as one of generalisation: systems receive English genre definitions and limited synthetic training data, but the hidden benchmark contains noisier naturally occurring Arabic spanning Modern Standard Arabic, Classical Arabic, and multiple dialects, with *previously unseen genre–definition combinations*. Because the 74 specific test genres are disjoint from training, memorisation is useless and semantic understanding of communicative function is required. The official ranking metric is

$$\text{Hier. F1} = \frac{1}{2}\left(\text{Broad F1}_{\text{macro}} + \text{Spec. F1}_{\text{macro}}\right).$$

Our contributions are:

1. A hierarchically constrained, definition-guided LLM judge that reaches 0.7139 Hierarchical Macro F1 with no task training.
2. An account of why one predictor fails the two tiers differently – broad accuracy is just the family-accuracy of the specific prediction, whereas Broad Macro F1 is set by minority-class false positives – so correction has to be tier-specific (§7).
3. A full-context multi-model consensus correction, applied conservatively (agreement-gated, direction-verified), with a three-arm ablation separating two causes of the topic trap: an explicit type-vs-topic instruction and a visible label to move the error onto, each accounting for about half of the residual.
4. An attractor-drain step targeting the macro-F1 failure mode of over-predicting generic classes.
5. Empirical validation on the hidden test: 0.7352 Hierarchical Macro F1, ranked 1st of 18 teams, each component scored as its own submission.

Code and artifacts: https://github.com/Hassn11q/thakaa-aragenre-2026. The main challenges were a specific-label inventory disjoint from training, a topic trap (writing *about* a subject vs. the subject's genre), and classes with no written signal (song dialects, textbook levels).

2 Related Work

Arabic and hierarchy-aware HTC. Arabic pretrained models (Antoun et al., 2020, 2021) underpin most classification work on the language, including hierarchical variants (Bouchiha et al., 2025, 2026), and a large literature models the label hierarchy explicitly (Zhu et al., 2023; Cai et al., 2024; Zeng et al., 2025; Wang et al., 2025; du Toit and Dunaiski, 2025; Zangari et al., 2024). These closed-label supervised setups do not transfer to AraGenre's unseen, definition-specified labels; we instead enforce the hierarchy structurally, forcing broad $=$ parent(specific).

Zero-shot HTC, LLM judging, and consensus. Prior systems reach an unseen label space through prompt ensembles (Xia et al., 2025), LLM-built prototypes (Zhang et al., 2025) or rewritten taxonomies (Golde et al., 2026); we keep the definitions verbatim and vary only how much of the taxonomy is visible at once. Our two-stage design retrieves candidate definitions (Chen et al., 2024) and compares them with an LLM (Sun et al., 2023; Zheng et al., 2023), with Qwen3-Embedding (Qwen Team, 2025) bridging Arabic and English; the chain-of-thought pass follows reasoning-based HTC (Jiang et al., 2025; Pan and Wu, 2025). Ensembling diverse predictors (Dietterich, 2000) appears in LLM form as juries (Verga et al., 2024), blending (Jiang et al., 2023), mixtures of agents (Wang et al., 2024), debate (Du et al., 2024) and self-consistency (Wang et al., 2023); those aggregate free-form text or resample one model, whereas we require two models to emit the *same* label, which helps only with the full taxonomy in context.

3 Background

Each instance is `{id, text}`; systems output one `broad_genre` (of 6: Informative, Creative, Interactive, Learning, Legal, Religious) and one `specific_genre` (of 74), using exact taxonomy label names, each with an English definition. The released data are tiny and synthetic-like (train: 140 items / 7 specific genres; dev: 110 / 6), and the specific-label inventories are *100% disjoint* across

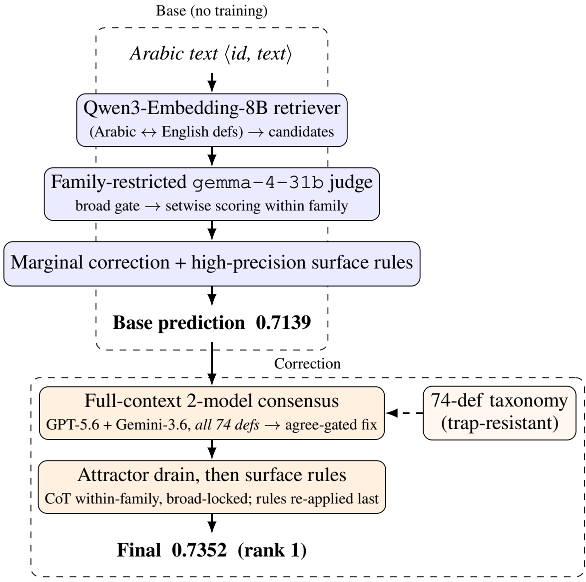

Figure 1: System overview. The base judge yields 0.7139; full-context two-model consensus plus a within-family attractor drain correct both tiers, reaching 0.7352 (rank 1).

train, dev and test, so no supervised label set transfers. The *hidden test* has 27,972 naturally occurring instances over 74 specific genres in 6 families: Informative (42), Learning (12), Creative (7), Religious (6), Interactive (4) and Legal (3). Informative is dominated by 20 job domains and 13 book-description topics; Creative holds 5 song dialects. Missing predictions are counted incorrect; the stronger official dev baseline is in Table 1. Informative dominates our predictions ($\sim$50%), and the specific tier's 74 classes make Macro F1 highly sensitive to rare-class recall (e.g., dialect song lyrics, niche job domains).

4 System

4.1 Base: family-restricted definition-guided judge

The base system turns the 74-way choice into a within-family one (Figure 1): retrieval proposes candidates and a judge scores only siblings of a gated broad genre. The setwise judge is `google/gemma-4-31B-it` (Gemma Team, Google DeepMind, 2026), served through vLLM under the local alias `gemma-4-31b`; API model strings are pinned in Appendix H. Each text passes through four steps. (i) `Qwen3-Embedding-8B` (Qwen Team, 2025) encodes the Arabic text and the English definitions, and proposes candidate specific genres by cosine similarity. (ii) Decoding is hierarchically constrained, all siblings for small families and top-k retrieved for large topical ones, so that broad $=$ parent(specific) and broad *accuracy* equals the family-accuracy of the specific prediction. Broad Macro F1 does not: it averages per-class F1, and the gap between the two is what the diagnosis below exploits. (iii) The judge scores each candidate 0–100 against its definition in one setwise call (Sun et al., 2023), and we take the argmax after a light marginal correction (Appendix G). (iv) A few high-precision surface rules override the judge only on near-unambiguous markers (Qur'anic mushaf orthography $\rightarrow$ `quran`; isnād openings $\rightarrow$ `hadith`; we screened samples by hand and saw no false positives, though without gold this is an impression, not a precision estimate; emoji/hashtag short posts $\rightarrow$ Interactive subtypes). The judge also sees 74 hand-written per-genre contrastive discriminators appended to the definitions; hyperparameters are in Appendix H. On the hidden test this base pipeline scores 0.7139.

Diagnosis from the scoreboard. The leaderboard exposed the structure: against the system then ahead of us, our base led on Specific Macro F1 (0.5487 vs. 0.5339) but trailed on Broad Macro F1 (0.8792 vs. 0.90), so the whole deficit sat in the broad tier. Broad accuracy (0.9003) exceeded broad macro, indicating uneven per-class performance; reading the errors showed cross-family false positives, with Bible passages filed as poetry or encyclopedia entries and news filed as legal.

4.2 Correction: full-context multi-model consensus

We re-predict the specific genre with two frontier instruction models (GPT-5.6 Luna (OpenAI, 2026) and Gemini-3.6 Flash (Google, 2026)), *feeding each the entire 74-definition taxonomy* ($\approx$2.9K tokens) grouped by broad genre, and asking for the single best specific label. This follows the ensemble argument that combining diverse predictors beats any single one (Dietterich, 2000), in its LLM form (Verga et al., 2024; Wang et al., 2024); we claim model diversity, not statistical independence. (Claude Sonnet 5 (Anthropic, 2026) was run on the diagnostic pool but excluded from the deployed consensus on cost grounds.)

What suppresses the topic trap. Given only the 6 broad definitions, all three verification models fell into a *topic trap* on the book descriptions: text *about* Islam $\rightarrow$ Religious, even when it was an Informative *book description*. The full-taxonomy prompt also carries an explicit type-vs-topic instruction, so we ran a three-arm ablation over the same 1,403 instances to estimate each factor's incremental effect (one model, GPT-5.6, identical pool): with broad definitions only and no such instruction, 394 are called Religious; adding the instruction alone removes 41% of those (394$\rightarrow$233); supplying the full taxonomy removes a further 49% (233$\rightarrow$119). The two are not interchangeable: the instruction helps only on the book descriptions and over-fires elsewhere, whereas the taxonomy improves every subset of the pool (Table 3). Scripture filed as an encyclopedia entry (هؤلاء امراء بني عيسو, Genesis 36) is moved to `bible`, and base-family agreement rises in five of the six families (Appendix E).

Conservative application. The first model runs over all 27,972 instances, the second only over the 8,041 where the first disagrees with the base, and a correction is applied only where both name the same label. Broad-changing moves are gated to seven read-verified directions, led by Informative$\rightarrow$Interactive for comments and Legal$\rightarrow$Informative for wire news (Appendix I); unreliable directions (e.g. Informative$\rightarrow$Learning, which over-fires student-writing on academic prose) are rejected, 718 moves in all. Within-family refinements are applied wherever both models agree.

4.3 Attractor drain: recovering starved rare classes

Reading our own hidden-test predictions during the evaluation phase revealed five large "attractor" classes that absorb texts the judge cannot place, starving rare siblings. They come from that prediction distribution, not from held-out gold. The drain is deterministic: when the base label is an attractor class and a separate chain-of-thought rejudging pass (Wei et al., 2022) names a different sibling in the same family, the sibling replaces it (broad never changes). The surface rules are then re-applied over the corrected labels, acting as a guard rather than a gain (Appendix I). On the base alone it moves 554 instances, 6–35% of each attractor class (Appendix F); in the final pipeline consensus runs first, so the drain fires on 383. Relative to the base, the final system changes 459 broad and 2,890 specific labels (10.3% of instances).

System	Split	Hier. F1
Official baseline (embedding sim.)	dev	0.4658
Our judge (Gemma-4-31B)	dev	0.8794
Our judge (GPT-5.6)	dev	0.8943
Gemma-4-31B + consensus	dev	0.8557
GPT-5.6 judge + attractor drain†	dev	0.9358
Genre-general zero-shot judge	test	0.6067
+ embedding retrieval gate	test	0.6812
family-restricted, derived broad	test	0.6882
+ rules / calibration (base)	test	**0.7139**
+ 2-model consensus (pool)	test	0.7224
+ safe broad expansion (full set)	test	0.7256
+ full consensus + drain (final)	test	**0.7352**

Table 1: Main progression on both splits. Dev is near-degenerate (its 6 specific genres map one-to-one onto 6 broad, so all three F1 metrics coincide), so the hidden test is our primary evaluation. †applied to the GPT-5.6 judge, tuned on dev, so not held out.

System	Hier	SpM	SpW	SpA	BrM	BrW	BrA
Base (ours)	.714	.549	.584	.591	.879	.900	.900
+ consensus	.722	.553	.592	.598	.892	.910	.910
Final (rank 1)	**.735**	**.573**	**.613**	**.619**	**.898**	**.914**	**.914**

Table 2: Official metrics for our three scored systems on the hidden test, rounded to three decimals (Sp/Br = Specific/Broad; M/W/A = Macro F1, Weighted F1, Accuracy). Consensus on the verification pool lifts the broad tier by +.013 Broad Macro F1; the final row folds in the safe broad expansion, consensus over the full set and the drain, together worth +.020 Specific Macro F1.

5 Experimental Setup

Data splits. We train nothing; every system is applied unchanged to the hidden test. An input is `{"id","text"}` with Arabic text such as مطلوب سائق باص 24 راكب في شركة في دبي جبل علي (test id 24486, "bus driver wanted, 24-seater, for a company in Dubai, Jebel Ali"); the required output pairs it with `broad_genre` = Informative and `specific_genre` = `drivers_and_delivery_jobs`.

Preprocessing. None. Normalisation erases the two orthographic signals that separate whole families here: heavy diacritisation marks 77% of the texts we label `quran` and 43% of poetry against 1% elsewhere, and emoji mark 28% of Interactive texts against under 1% elsewhere (Appendix I).

Tools and metrics. Qwen3-Embedding-8B under `sentence-transformers`, vLLM for the local judge, and the vendor SDKs for the verifiers; we report Macro F1, Weighted F1 and Accuracy at both tiers. Versions, URLs and hyperparameters are in Appendix H.

Cost. Consensus makes 36,013 calls at a ≈3.0K-token prompt each, about US$50; rates and timings are in Appendix H. The base needs no paid API.

6 Results

6.1 Development-set results and component transfer

Re-running the pipeline on dev (110 examples, official gold) separates what transfers from what does not. The drain transfers: no class list is carried over, and the two dev classes it fires on are the two most-predicted there, so no gold is needed to find them. It moves five labels for +0.042 (paired bootstrap over the 110 dev items, 10k resamples: 95% CI [+0.010, +0.082]). The consensus stage does not: dev has no `*_book_description` and no scripture, so the trap it repairs has zero instances; two corrections fire and both are wrong. Residual dev errors stem from a convention clash between splits (Appendix B).

7 Analysis

Broad Macro is dominated by minority-class false positives, Specific Macro by rare-class recall, so gains on both compound through the mean. Broad-only prompting put 157 book descriptions under Religious unanimously across all three models, which the full taxonomy largely reversed; and a single model disagreed with the base on ∼29% of specifics, too noisy to act on without a second. Every alternative we tried for the correction stage fell below the 0.7139 base; the scores are in Appendix D.

Error analysis (hidden test). Three types dominate: cross-family topic leaks, attractor overprediction, and ceiling classes where 87–99% of the texts we label as song lyrics carry no marker from that dialect's cue list. Examples in Appendix C.

8 Conclusion

Training-free hierarchical Arabic genre classification is achievable: a constrained definition-guided judge, an agreement-gated consensus and a rare-class attractor drain ranked 1st of 18 at 0.7352 Hierarchical Macro F1. The taxonomy matters because it gives the model a label to move an error *to*.

Deriving the move directions and the attractor list from prediction statistics rather than from reading outputs, and finding written correlates for spoken dialect—our clearest ceiling—are the next steps.

Limitations

What is and is not a score. Every hidden-test *score* we report is an official evaluation-phase submission (Appendix I); we made no submission after the deadline. The instruction-only arm of the three-arm ablation (§4) was run after the phase closed, so that comparison is post hoc, and the descriptive tables in Appendices C, E, F and I are local analyses of submitted outputs rather than scored runs. **Validation.** The hidden test has no released gold and the development set is near-degenerate, so components were selected on the public leaderboard and by manual reading; our reported progression therefore doubles as the selection signal, which risks fitting the test distribution. Because blind labels are unreleased, no hidden-test gain carries a confidence interval, and those gains should be read as leaderboard-measured rather than statistically certified. The one component testable against gold is the drain, where a paired bootstrap on dev puts its gain above zero (95% CI $[+0.010, +0.082]$). **An untested comparison.** We ran GPT-5.6 over the full test set but never submitted its predictions as a system, so the claim that a cheap base judge with conservative correction beats direct full-taxonomy prediction is untested; the $\sim$29% disagreement rate is our only evidence that one model is too noisy to act on. **Cost and ceilings.** Consensus cost scales with test size, and song-dialect distinctions remain near the noise floor because the distinguishing features are phonetic and largely absent from written lyrics.

Ethics Statement

This work uses only the publicly released AraGenre data and collects nothing new; any user-generated text originates from the official benchmark. In line with the shared-task rules, all predictions are produced automatically: no submitted label was hand-edited and we did not reverse-engineer the hidden evaluation labels. We did read system outputs and the corresponding texts to design the automatic rules, and the rule-precision estimates in §4 come from that inspection; this is author inspection of system outputs, not annotation of the evaluation set, and no such judgement enters a submission except through a deterministic rule.

The taxonomy spans religiously and culturally sensitive categories (Islamic and Christian scripture, dialectal varieties, regional song lyrics), which we treat descriptively as text-type labels: a misclassification carries no endorsement and no judgement on the authenticity, orthodoxy or quality of a text, and models trained on skewed corpora may under-serve lower-resourced dialects. Two misuses would concern us. Dialect is not provenance – pointed at ordinary prose, the classifier reads as a guess at where a writer is from, and our own measurements show that the written signal is mostly absent. And because the labels separate religious from informative text, genre output could be used to route or suppress material by register rather than by content. The system assigns text types, it does not assess them, and a 0.57 Specific Macro F1 is not a basis for an automated decision about a document or its author without a person in the loop.

Our correction stage sends Arabic text to third-party hosted LLM APIs; the inputs are already-public benchmark data, but the same pipeline should not be applied to private or sensitive text without review. Reliance on hosted models whose availability, cost and behaviour may change limits equitable reproducibility; the base retriever-plus-open-judge pipeline is provided as the self-hostable alternative.

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A Reproducibility

Official submission. The results we report are those of submission 872515, evaluated 1 August 2026, which scored 0.7352 Hierarchical Macro F1 (Specific Macro 0.5725, Weighted 0.6125, Accuracy 0.6189; Broad Macro 0.8979, Weighted 0.9142, Accuracy 0.9138) over 27,972 of 27,972 gold instances with no missing prediction, ranking 1st of the 18 teams in the organisers' final standings, which we transcribe in the repository for reference. Every hidden-test number in Table 1 and Appendix D is likewise an official scored submission rather than a local estimate; the submission IDs are listed in Appendix I. All inference is zero-shot with no task-specific training.

Data: we use only the official AraGenre inputs and definition files, and no external labelled data. The consensus verifiers receive the 74 definitions verbatim. The judge does not: by default it appends a one-line Arabic discriminator to each definition and prepends per-family cue lists, both author-written from the released definitions and Arabic linguistic knowledge, and both naming genres explicitly (`src/discriminators.py`, `src/judge.py`). The surface rules name a further seven labels. No test label or annotation informs any of it, but the pipeline is not label-agnostic and we do not claim it is.

Models: the base retriever is Qwen3-Embedding-8B (4-bit NF4 weights with fp16 compute, 4096-dim, cosine over L2-normalised vectors, sequence length capped at 512 tokens); the judge is a Gemma-family model (`gemma-4-31b`) served via vLLM with setwise 0–100 scoring; the consensus verifiers are GPT-5.6 Luna and Gemini-3.6 Flash, each queried with the full-taxonomy prompt.

Settings: the judge decodes at temperature 0 with seed 42; the Gemini verifier runs at temperature 0 and the GPT verifier at `reasoning_effort=low` with no temperature set. The retriever caches the top 20 specific genres per text; the judge scores every sibling in the four small families and the top 6 retrieved elsewhere, in a single setwise pass, with marginal correction α=0.5.

Pipeline and code: paths below are relative to the repository root. Every stage is released, one file each: retrieval (`src/retrieve.py`), the family-restricted judge (`src/judge.py`), the surface rules (`src/rules.py`), the full-context verifiers (`src/verify.py`), the consensus and drain assembly (`src/assemble.py`), and the submission builder (`src/submission.py`), which enforces broad=parent(specific) and validates that all 27,972 IDs are present with exact labels.

What the release cannot regenerate. Two inputs to the trap experiment have no producer here: the 2,635-instance verification pool, selected during the evaluation phase as the instances whose broad label we judged least secure, and the broad-only prompt behind arm A. The pool file and arm A's cached outputs ship, so `src/trap_table.py` rebuilds the table offline, but neither selection can be re-derived from the code.

Two cached inputs. The broad gate the judge restricts to is submission 866538 (0.6812), an earlier retrieval-gated system we submitted and scored, reused verbatim rather than regenerated by the released code; and the base predictions additionally carry an Interactive-recall reassignment applied during the evaluation phase by a separate automatic judge pass (`src/interactive_judge.py`), which we release but which is not wired into `reproduce.sh`; re-running the released judge over the same candidate sets reproduces 93.5% of the base specific labels and 97.1% of the broad labels. Both files ship in `artifacts/`, so the submitted predictions rebuild exactly (27,972/27,972) from cached outputs, but the base stage is a cached input rather than something the repository regenerates end to end.

Caveats: the hosted API models are versioned by release date and are not guaranteed bit-reproducible across dates; the consensus stage incurs API cost proportional to the number of verified instances (on the order of tens of US dollars for this test set). The base pipeline runs on a single CUDA GPU (embeddings) plus a remote vLLM endpoint (judge).

B Annotation-Convention Divergence Between Splits

Five of the seven residual dev errors are texts that analyse religion from the outside, e.g. تشير بعض الدراسات إلى أن أساليب الوعظ تختلف بين المجتمعات ("some studies indicate that preaching styles differ between societies"). Our system labels these `Analytical Reports` (Informative); the dev gold label is `Religious Commentary` (Religious).

This is the *inverse* of the convention in the hidden test, where `islamic_book_description` ("texts summarising Islamic or religious books") is Informative and our consensus stage is rewarded for moving such texts out of Religious. Both conventions are stated in their own definition files, and the two files disagree about where writing *about* religion belongs. A system that follows the test convention is therefore necessarily penalised on dev, which bounds how far dev accuracy can be pushed without split-specific rules. It also supports our central claim: the convention is carried by the definitions, which is why supplying the full taxonomy, rather than the broad labels alone, changes model behaviour so sharply.

C Error Analysis Examples

Three error types dominate. (i) *Cross-family topic leaks*: a description *of* a religious book (يهدف هذا الكتاب إلى قراءة سيرة الرسول, id 10041) is `islamic_book_description` (Informative), which our base and final systems both get right but which all three verifiers call Religious under the broad-only prompt; scripture leaks the other way (هؤلاء امراء بني عيسو, Genesis 36, id 10798), where the base says `history_encyclopedia` and consensus corrects it to `bible`. (ii) *At-*

tractor over-prediction: long prose absorbed into `egyptian_song_lyrics`. (iii) *Ceiling classes*: for each of the five song-lyric classes, 87–99% of the texts we assign to it contain no marker from that dialect's cue list (`src/dialect_markers.py` ships the lists and reproduces the table; the rate is a property of the text, so it is measurable without gold).

D Negative Results

A *direct* six-way broad classifier regressed to 0.6206, confirming that deriving broad from the specific prediction beats classifying it directly; scoring all 74 candidates without retrieval pruning fell to 0.6517. For correction, plain self-consistency (0.7134), entropic optimal transport (0.7115), uniform-prior calibration (0.7124) and an unrestricted chain-of-thought pass (0.7131) all regressed below the 0.7139 base, whereas full-context consensus improved it. Cross-lingual rerankers failed (Aarab, 2026), regex rules added nothing the judge lacked, and the off-the-shelf dialect classifiers we tried did not expose the Iraqi/Sudanese distinction that NADI-style country-level tasks define (Abdul-Mageed et al., 2021).

E Full-Context vs. Broad-Only Prompting

Both consensus models were run over the same verification pool ($n = 2{,}635$) once with only the 6 broad definitions and once with all 74 specific definitions. Because the full-taxonomy prompt also carries an explicit type-vs-topic instruction, we added a third arm, broad definitions only *plus* that instruction, to separate the two factors. On the 1,403 book-description instances GPT-5.6 calls 394 of them Religious with neither, 233 with the instruction alone and 119 with the full taxonomy; `src/trap_table.py` rebuilds these counts from cached verifier outputs with no API access.

Table 3 splits the same three arms over the whole pool. Two things follow. First, the reduction is not a prior shift: from A to C the verifier calls Religious *more* often on the instances the base already reads as Religious and less often on every other subset, so the separation between the two widens monotonically. Second, the instruction and the taxonomy are not interchangeable. The instruction pays off only on the book descriptions, where the competing label (`*_book_description`) is the trap's target; on the rest of the pool it over-fires, raising Religious calls by a quarter. Only the full taxonomy improves every subset at once. We read this as the instruction supplying a convention the model cannot act on until the taxonomy shows it a label that satisfies the convention. The caveat of Table 4 applies here too: the row labels come from the base system, so these are agreement counts, not accuracy.

Religious calls	A	B	C
book descriptions (n=1,403)	394	233	**119**
rest of pool (n=1,169)	121	152	**61**
base-Religious (n=63)	26	28	**30**
separation (pp)	21	29	**41**

Table 3: The three arms over the full verification pool (GPT-5.6). A: broad definitions only. B: + type-vs-topic instruction. C: + full 74-definition taxonomy. The first two rows should fall and the third should rise; only C does both. Separation is the Religious-call rate on base-Religious instances minus the rate on the other 2,572. Rebuilt by `src/trap_table.py`.

Table 4 reports how often GPT-5.6's broad verdict still agrees with the base system's family, one model across the two original prompts. Broad-only prompting collapses on families whose members are *about* another family's subject matter (Learning, Informative), which is exactly the topic trap; the full taxonomy repairs it. Requiring both verifiers to agree, joint Religious assignments on the same 1,403 instances fall from 235 under broad-only prompting to 32 under the full taxonomy, an 86% reduction.

Base family	n	broad-only	full-taxonomy
Informative	1682	49%	**78%**
Learning	305	18%	**74%**
Interactive	153	46%	**72%**
Legal	281	53%	**56%**
Creative	151	50%	48%
Religious	63	41%	**48%**

Table 4: Share of instances where GPT-5.6's broad verdict matches the base family, by prompt context. This measures *agreement with the base system*, not accuracy: the hidden test has no released gold, so we can show that full context changes verifier behaviour and that the change is consistent with the taxonomy's own conventions, but we cannot certify it as an accuracy gain. The dev contrast in §6.1 is the gold-anchored counterpart.

F Attractor Drain

The drain is deterministic: it fires when the base label is an attractor class and the chain-of-thought pass names a different sibling inside the same broad family (Table 5). No score threshold or entropy criterion is used, and broad is held fixed by construction.

Class	base	moved
`literature_fiction_bk_desc.`	598	208 (35%)
`egyptian_song_lyrics`	979	212 (22%)
`msa_poetry`	608	52 (9%)
`culture_encyclopedia`	455	41 (9%)
`history_encyclopedia`	632	41 (6%)

Table 5: Attractor classes and how many predictions the drain reassigns to within-family siblings when applied to the base predictions alone. In the submitted pipeline consensus is applied first and the drain fires on 383 of these instances.

G Calibration

The correction subtracts a scaled per-genre mean from each candidate's score, $s'(y) = s(y) - \alpha\,\bar{s}(y)$, where $\bar{s}(y)$ is the mean score genre y received over the whole test set and α=0.5. It damps genres the judge rates generously everywhere, which is what starves rare siblings. We started from α=0.75, chosen on the released training labels; it over-flattened once the label space grew from 7 to 74 classes. The correction is applied before the argmax and never changes the broad family. Two alternatives we tried instead of it, uniform-prior calibration and entropic optimal transport, both scored below the base (Appendix D).

H Hyperparameters and Prompts

Retriever: Qwen3-Embedding-8B loaded in 4-bit NF4 with fp16 compute (the configuration the submitted run used; only the cosine ranking is consumed and NF4 preserves embedding direction), 4096-dim, L2-normalised, `max_seq_length` 512, the top 20 specific genres cached per text; the judge scores every sibling in the four small families (Creative 7, Religious 6, Interactive 4, Legal 3) and the top 6 retrieved in-family candidates in the two large ones.

Judge. `gemma-4-31b` via vLLM, setwise 0–100 scoring in a single pass, temperature 0, seed 42, 3 retries with JSON-regex fallback, marginal correction α=0.5.

Consensus verifiers. GPT-5.6 Luna (`reasoning_effort=low`, ≤500 completion tokens) and Gemini-3.6 Flash (temperature 0, ≤400 output tokens), each given the full 74-definition taxonomy (2,884 tokens under `o200k_base`, giving a mean prompt of 3,034 tokens and a maximum of 5,185) in the system prompt, and the text truncated to 1500 characters. The judge prompt asks for a JSON array of integer scores over the candidate definitions; the consensus prompt asks for a single exact `specific_genre` identifier and states that a factual description *of* a book is a `*_book_description`, not the book's subject genre.

Software. `torch` ≥2.1, `transformers` ≥4.44 (`https://github.com/huggingface/transformers`), `sentence-transformers` ≥3.0 (`https://github.com/UKPLab/sentence-transformers`), `bitsandbytes` ≥0.43 (`https://github.com/bitsandbytes-foundation/bitsandbytes`), vLLM (`https://github.com/vllm-project/vllm`), `openai` ≥1.99 (`https://github.com/openai/openai-python`) and `google-genai` ≥0.3 (`https://github.com/googleapis/python-genai`). Model weights: Qwen3-Embedding-8B (`https://huggingface.co/Qwen/Qwen3-Embedding-8B`).

Cost. At list prices the consensus stage runs about US$1.05–1.48 per 1,000 calls, giving the $50 total quoted in §5 over its 36,013 calls.

Throughput. A consensus call takes 2.0–2.8 s. At the released default of 20 concurrent workers the 8,041-instance Gemini pass took 19 min of wall clock; the full GPT pass over 27,972 instances was run at higher concurrency.

I Official Submission Record

Every hidden-test number we report is an official evaluation of a submitted file (Table 6), so the progression is traceable to scored submissions rather than to local approximations.

Two readings this record supports. Submission 867465 isolates the surface rules on top of 867323: they rewrite 32 labels (23 `quran`, 9 `hadith`) and leave the score at 0.6881 against

ID	System	Hier. F1
865908	genre-general zero-shot judge	0.6067
866538	+ embedding retrieval gate	0.6812
867323	family-restricted, derived broad	0.6882
867376	direct six-way broad judge	0.6206
867451	unpruned 74-way scoring	0.6517
867465	+ surface rules only	0.6881
870640	base system	0.7139
870841	uniform-prior calibration	0.7124
870979	self-consistency (K=3)	0.7134
870981	entropic optimal transport	0.7115
871127	chain-of-thought applied wholesale	0.7131
871609	+ 2-model consensus (pool)	0.7224
872511	+ safe broad expansion	0.7256
872515	**+ full consensus + drain**	**0.7352**

Table 6: Official submissions behind Table 1 and Appendix D. Submission 872515 is the final system: 27,972 predictions over 27,972 gold instances, none missing. Submissions were made sequentially over four days, so the progression is a record of scored configurations rather than a single controlled sweep.

0.6882. The rules are precise but score-neutral at that point, so the 0.6882→0.7139 gain belongs to the judge and calibration changes, not to them. They were kept for a different job: in the final pipeline they fire on 1,087 instances and agree with the base label on every one, so their whole net effect is to restore 75 labels that consensus or the drain had moved away (56 `youtube_comments`, 19 `quran`). They guard the correction stages rather than adding score. Submission 871127 applies the chain-of-thought pass to every instance and scores 0.7131, below the 0.7139 base; the same pass restricted to attractor classes is what the drain uses. Chain-of-thought helps here only where it is aimed.

Gated broad moves. Of the 3,300 instances where both verifiers agree against the base, 2,122 stay inside the base family and are applied as refinements. Of the 1,178 that would cross families, 460 fall in a read-verified direction and are applied: Informative→Interactive 195, Legal→Informative 80, Creative→Religious 58, Informative→Religious 58, Creative→Informative 44, Interactive→Religious 20, Legal→Interactive 5. The remaining 718 are rejected, the largest groups being Informative→Learning (219), Creative→Interactive (122) and Learning→Informative (99).

Preserved rule defects. Three regexes in `src/rules.py` do not match their intent: the Qur'anic class also admits U+0671, the opinion class lets a bare variation selector match, and one character class contains a literal `|`. They ran as written when the submission was produced, so we keep them verbatim and document them in the code; correcting all three changes 22 of the 27,972 final predictions.

Orthographic signals. Share of each group carrying heavy diacritisation (a mark on over 10% of its Arabic characters) or at least one emoji, over the final predictions: `quran` 77%/0% (n=691), poetry 43%/0% (n=1,606), `hadith` 1%/1% (n=647), book descriptions 1%/0% (n=3,916), the whole Interactive family 0%/28% (n=1,978; `twitter_posts` 57%, `youtube_comments` 30%, `app_reviews` 11%, `book_reviews` 0%). Both are properties of the text, so no gold is needed; `src/orthographic_signals.py` reproduces the table.